

$$(h_i = \frac{\lambda_i}{2})$$
$$(h_i = \frac{\lambda_i}{2})$$

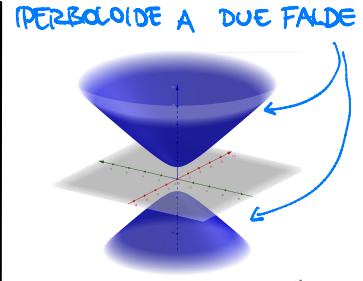
VON A CENTRO
CENTRO

$n=3$ SUPERFICI QUADRICHE



$$h_1, h_2 > 0, h_3 < 0$$

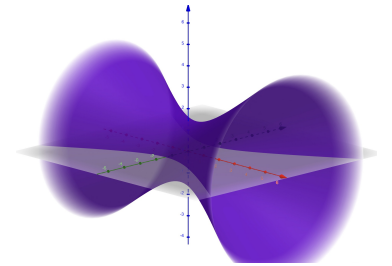
$$x^2 + y^2 - z^2 + 1 = 0$$



$$h_1 > 0, h_2, h_3 < 0$$

$$x^2 - y^2 - z^2 + 1 = 0$$

IPERBOLOIDE AD UNA FALDA



$$h_1 x^2 + h_2 y^2 + h_3 z^2 = 0$$

$$h_1, h_2, h_3 > 0 \quad x^2 + y^2 + z^2 = 0$$

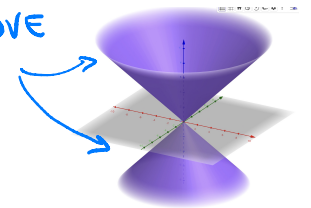
$$h_1, h_2, h_3 < 0 \quad x^2 + y^2 + z^2 = 0$$

PUNTO (0,0,0) DOPPIO

UNO POSITIVO (NEGATIVO) E GLI
ALTRI NEGATIVI (POSITIVI)

$$x^2 + y^2 = z^2$$

CONO A DUE
FALDE



$$h_1, h_2 > 0 \quad x^2 + y^2 + 1 = 0$$

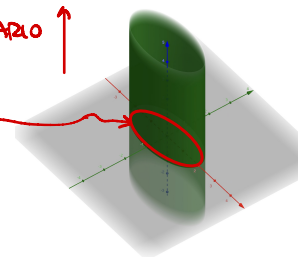
CILINDRO IMMAGINARIO

CILINDRO ELLITTICO

$$h_1 x^2 + h_2 y^2 + 1 = 0$$

$$h_1, h_2 < 0 \quad -x^2 - y^2 + 1 = 0$$

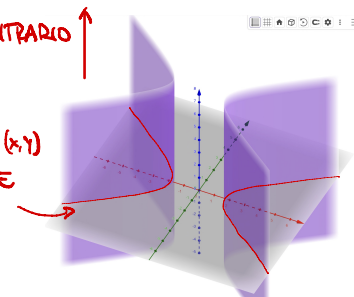
z ARBITRARIO
SEZIONE (x,y)
ELLISSE



$$h_1 > 0, h_2 < 0 \quad x^2 - y^2 + 1 = 0$$

CILINDRO IPERBOLICO

z ARBITRARIO
SEZIONE (x,y)
IPERBOLE



$$h_1, h_2 > 0 \quad x^2 + y^2 = 0$$

$$h_1, h_2 < 0 \quad x^2 + y^2 = 0$$

RETTA DOPPIA (0,0,z)



$$h_1 x^2 + h_2 y^2 = 0$$

$$h_1 > 0, h_2 < 0$$

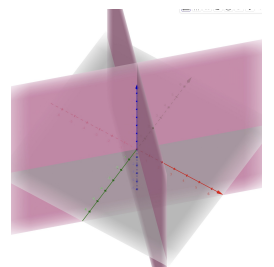
(VICEVERSA)

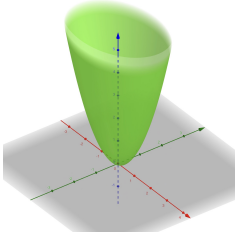
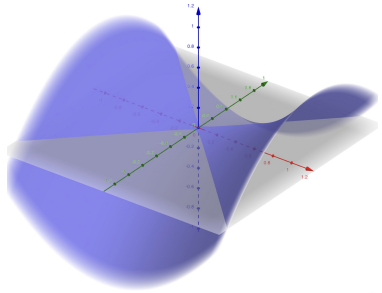
$$x^2 - y^2 = 0$$

PIANI REALI INCIDENTI

$$(x, x, z)$$

$$(x, -x, z)$$



QUADRICE A NON A CENTRO	FORMA CANONICA METRICA	FORMA CANONICA AFFINE	TIPO AFFINE
	$h_1 x^2 + 1 = 0$	$h_1 > 0 \quad x^2 + 1 = 0$ $h_1 < 0 \quad x^2 = 1$	PIANI COMPLESSI // $(x = \pm i)$ PIANI REALI // $x = \pm 1$
	$x^2 = 0$	$x^2 = 0$	PIANO DOPPIO $x = 0$
	$h_1 x^2 + h_2 y^2 + 2 \ b_0\ z = 0$	$h_1, h_2 > 0$ $x^2 + y^2 + 2z = 0$	PARABOLOIDE ELLITTICO 
		$h_1 > 0, h_2 < 0$ $h_1 < 0, h_2 > 0$ $x^2 - y^2 + 2z = 0$	PARABOLOIDE IPEZBOLICO 
	$h_1 x^2 + 2 \ b_0\ z = 0$	$h_1 > 0$ $h_2 < 0 \quad x^2 - 2z = 0$	CILINDRO PARABOLICO 